Reinforcement Learning for Ambulance Dispatch and Repositioning

Anuj Trivedi (B23CH1006)

Indian Institute of Technology Jodhpur — AI Mini Project, Assignment 2

Repository: https://github.com/anujtrivedi06/ambulance-dispatch-repositioning

1. Problem and MDP Formulation

Emergency calls arrive at random locations and times across a city road network. A fleet of K ambulances must answer them, and an agent makes two coupled decisions: **dispatch** (which unit to send to an incoming call) and **repositioning** (where each idle unit should wait). The two conflict. Sending the nearest unit is optimal for the call in hand but can strip cover from a dense district, costing the next call more than it saved this one.

Decisions occur only at irregular event epochs — a call arriving, or a unit becoming free — so the problem is a **semi-Markov decision process** rather than a fixed-step MDP. Transitions carry an elapsed time Δt, and discounting must respect it.

2. State Space (S)

The underlying state is $s = (t, \{(status_k, loc_k, eta_k)\}_{k=1..K}, queue)$: the clock, every unit's status, position and arrival estimate, and the calls still waiting. With |V| = 64 nodes, six unit statuses and K = 6 ambulances, that is on the order of $(64 \times 6)^6 \approx 10^{15}$ configurations before the continuous clock is counted. Exact dynamic programming is impossible and function approximation is mandatory.

The agent therefore observes an **11-dimensional post-decision feature vector**: available-unit fraction; time of day as (sin, cos); expected covered demand; worst-zone coverage; demand-weighted distance to the nearest free unit; four per-zone coverage values; and a bias term. The post-decision state is a deterministic function of (state, action), so the agent can rank every candidate action by one forward pass each, with no sampling.

The feature set was cut from twelve after diagnosis on collected data: three features were exactly redundant (design matrix rank 9 of 12) and the per-zone coverages were collinear at 0.97 and 1.00. Ridge hid this — value *prediction* stayed accurate, because collinear directions cancel in-distribution, while the greedy policy degraded, because an action moves along exactly the unidentified direction.

3. Action Space (A)

The action space is **discrete**. At a repositioning epoch the agent chooses one of M stations, |A| = M (M = 4 in the main configuration, swept to 8). At a dispatch epoch it chooses an available unit or defers the call to the queue. The learned policy acts on the repositioning choice, which is where the combinatorial difficulty lies; dispatch follows the nearest-available rule, held fixed across every method so that measured differences are attributable to repositioning alone.

4. Reward Function (R)

Reward accrues between decision epochs from the calls *reached* in that interval, which is when an outcome is determined:

$$r_k = \Sigma_{i \in reached[t_k,\, t_{k+1})} \rho(i)$$

with two incentive designs compared. The **binary** design $\rho(i) = 1\{T_i \le \tau\}$, τ = 8 min, is the operational EMS standard. The **shaped** design $\rho(i) = -min(T_i, C)/10$, C = 40 min, is continuous in response time.

Because epochs are irregularly spaced, discounting is continuous-time: $\gamma_k = exp(-\beta\Delta t)$ with β = 1/240 per minute. A flat per-decision discount would value a decision followed by a quiet night identically to one followed by a busy minute.

This choice dominated every other design decision in the project. The binary indicator cannot distinguish a 3-minute response from a 7-minute one, so it starves learning of gradient over most of the space it cares about. Switching to the shaped reward is worth +0.052 coverage for the linear agent and +0.148 for the neural one — larger than the difference between linear and neural function classes.

5. Environment and Simulation Dynamics

The environment is a purpose-built discrete-event simulator (no third-party RL environment was used). Calls follow a non-homogeneous Poisson process sampled by thinning against a 24-hour demand profile, with locations drawn from a mixture of uniform background and Gaussian hotspots. Travel follows shortest paths over a weighted road graph with multiplicative noise; on-scene durations are lognormal; 75% of calls require transport to hospital plus a 15-minute handover. The main configuration is an 8×8 grid city, 4 stations, 6 ambulances and 4 calls/hour, which holds fleet utilisation near 65% — loaded enough that poor repositioning costs response time, slack enough that good repositioning has somewhere to put units.

Every policy is evaluated on **identical seeded call streams**, so comparisons are paired and the arrival process cancels rather than drowning the policy effect. All reported differences use a paired t-test *and* a Wilcoxon signed-rank test over 30 replications of 7 simulated days, and are called significant only when both agree.

6. Algorithm

The agent is **neural fitted value iteration** over post-decision states — the DQN family, with the max-over-actions taken through the post-decision representation. Because the post-decision state is deterministic given (s, a), $\max_a Q(s,a) = \max_a V(s^a)$, so a single value head suffices and no per-station output is baked in — the same agent transfers to a city with a different number of stations.

Training alternates ε-greedy data collection under the current policy with regression onto TD targets $r + \gamma \max_{a'} V_{target}(s'^{a'})$, computed from a target network frozen once per round. The network is a 2×64 MLP (4,993 parameters) trained with Adam and weight decay. Collection stores every candidate action considered, not merely the one taken, since the target needs a max over actions the behaviour policy declined.

LSTD and **LSPI** were implemented over the same features for comparison, plus a cross-entropy policy search over the identical linear class as a control: it shares the hypothesis class exactly, so any gap between them is the training objective, not the representation.

Table 1. Headline results (30 paired reps × 7 days).

Policy	Cov.	Mean	Worst zone
Random repositioning	0.357	10.09	0.282
Nearest-idle, static base	0.765	5.82	0.727
MEXCLP stations	0.860	5.38	0.798
DMEXCLP (strong baseline)	0.872	5.26	0.822
Linear ADP, binary reward	0.809	5.78	0.573
Deep NFQ, binary reward	0.724	6.60	0.384
RL agent (NFQ, shaped)	**0.874**	**5.22**	0.822

7. Baseline Comparisons

Four baselines of increasing strength: random repositioning (a floor); nearest-idle dispatch with static home stations, which

is what most real services do; stations optimised offline by p-median and MEXCLP; and **DMEXCLP**, the dynamic redeployment heuristic from the EMS literature, which sends a freed unit to whichever station most increases expected covered demand. DMEXCLP is the hard target: greedy action selection on the coverage feature is algebraically identical to it, so the RL agent's hypothesis class provably *contains* the heuristic it is measured against. Losing to it is therefore a statement about the training signal, not about the features.

8. Results

Five independently seeded agents were trained for 8 rounds each. Each selects its checkpoint on a validation block disjoint from both training and test episodes; the submitted agent is the one with the best *validation* score, never the best test score. All figures are on the untouched test block (30 paired replications of 7 days). Reporting a single run would not be a result: initialisation and the ε-greedy data a run happens to collect can move the outcome by more than the effect being claimed.

Across the five seeds, test coverage is **0.875 ± 0.002** against the heuristic's 0.872, and **all five seeds beat it**. The margin is small and consistent rather than large: the submitted agent scores 0.874 coverage (+0.001, p = 0.47, not significant) but is **significantly better on mean response time** — 5.22 min against 5.26, a difference of −0.038 min (p = 0.002). Against the untrained floor it is +0.052 coverage ($p = 2\times10^{-17}$). With the binary reward the same architecture collapses to 0.724, and worst-zone coverage falls to 0.384: maximising a count of indicators is maximised by serving dense zones and abandoning sparse ones, an equity failure the pooled metric hides entirely.

The learning curve is flat. Validation coverage oscillates within a band of 0.019 across eight rounds with no upward trend, so additional training did not make the policy better — the agent reaches heuristic parity early and stays there. This is what the project's wider results predict: value-prediction error is dominated by exogenous call variation, so the directions an action can move the state contribute almost nothing to the objective being minimised, and the fit has little incentive to rank actions correctly. A cross-entropy control confirms the reading — optimising policy quality directly over the *same* linear class closes a gap that fitting the value function does not.

Generalisation is partial, and this is the agent's clearest weakness. On a demand profile with relocated hotspots, never seen in training, the agents hold their absolute performance (0.879 ± 0.005 across seeds) and stay far above the untrained floor, so they have not merely memorised one demand map. But the heuristic *improves* on that easier profile, to 0.887, and the agents do not follow: the advantage held on the training distribution is given up. The learned value function is tuned to the demand it saw, while the heuristic recomputes coverage from the live demand model and adapts for free.

How much was ever available, and why. Choosing well is worth +0.091 coverage over random parking, but the best and second-best stations differ by only about 10% of a single marginal gain, and sweeping load (4–6 calls/h) and station count (4–8) leaves that unchanged. The cause is the environment's geometry: the median trip between any two nodes is 7.5 minutes against an 8-minute target, so one station already reaches over half the demand and a second adds little the first did not cover. Tightening the target to 5 minutes raises the best-versus-second-best gap from 7% to 39%. **The negative results in this project should therefore be read as holding at a generous response target**; a tighter one is where search or learning has the most room to prove itself, and is the clearest next experiment.

9. Conclusions

Five findings, each measured rather than asserted. **(i)** Reward design dominated model class: binary versus shaped reward moved coverage by 0.148, more than any architectural choice tested. **(ii)** Value-prediction accuracy is not policy quality; a control sharing the hypothesis class exactly, but optimising the policy directly, performed better than the fitted value function. **(iii)** The learned policy reaches parity with a strong domain heuristic on the distribution it trained on, but gives that parity up under demand shift, where the heuristic's live recomputation of coverage adapts for free. **(iv)** Reporting that honestly — with confidence intervals, disjoint validation and test seeds, and two agreeing significance tests — is more useful than a tuned victory. **(v)** The ceiling on all of this is set by the environment: an 8-minute target on this city leaves the candidate actions nearly interchangeable, and tightening it is the experiment most likely to change the conclusions.

The repository contains the full environment, all agents, the trained network, 72 tests, and scripts reproducing every figure and table here.

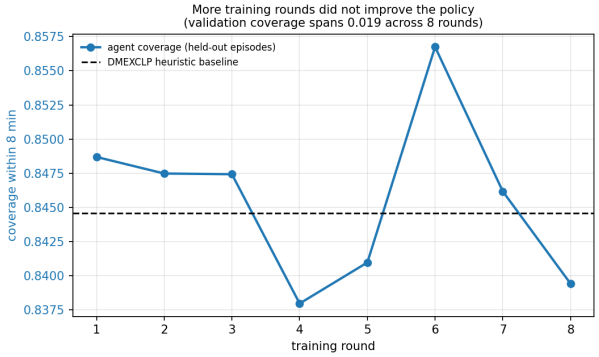


Fig. 1. Validation coverage over training rounds, against the heuristic baseline on the same episodes. Eight rounds produce no upward trend.

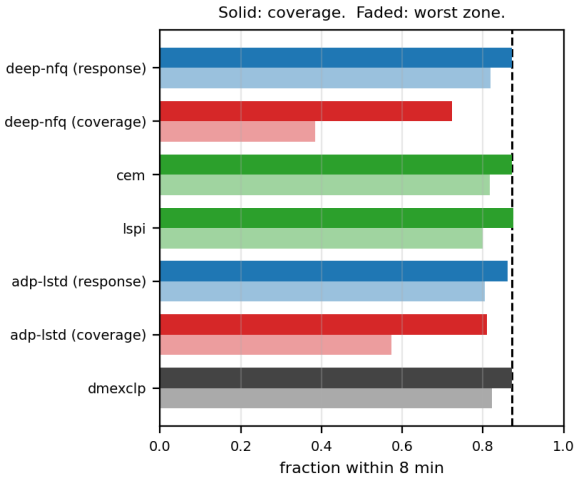


Fig. 2. Reward design dominates model class; the binary reward also collapses worst-zone coverage.